MATH 128a: Programming assignment #1

Instructor: F. Weber

Due by: **Thursday, September 25, 2025, 11:59pm**

Problem 1: Extrema

Write a program to take 6 numbers a, b, c, d, e, f and find the locations $x_{\min}$ and $x_{\max}$ of the absolute extrema of the function

$$p(x) = cx^3 + dx^2 + ex + f$$

over the interval $[a, b]$. Use the appropriate version of the quadratic formula (which depends on the parameters c,d,e) to avoid unnecessary cancellation of digits. Report the extrema returned by your code for the following choices of parameters:

trial	a	b	c	d	e	f
1	-1	2	-1	2	-1	1
2	1	2	1	-2	-1	1
3	-2	1	4	8	-4	-2
4	-1	2	1	0	1	-3
5	-0.3	0.6	10^{-14}	9	-3	0
6	-1	2	0	0	0	1.7
7	0	3	-3	9	-10^{-14}	0
8	0	1	0	-2	3	-1

Turn in your code and a table of the results of your code, $x_{\min}$, $x_{\max}$, $p(x_{\min})$, $p(x_{\max})$, reported to 6 digits of precision, and plots of $p(x)$ over $[a, b]$ for trials 3 and 7, with the extrema labeled. (You may annotate the hardcopy of the plots by hand if you wish.)

Problem 2: Sequences

Write a code that takes an integer, n, and returns the nth term in the following sequence:

$$a_1 = 1, \quad a_2 = \sqrt{1+2}, \quad a_3 = \sqrt{1+2\sqrt{1+3}}, \quad a_4 = \sqrt{1+2\sqrt{1+3\sqrt{1+4}}}, \quad \ldots$$

Evaluate a_n for $1 \le n \le 40$. Guess the limiting value of the sequence, $a = \lim_{n\to\infty} a_n$, and make a plot of $\ln(|a_n - a|)$ vs n. Also plot the line $y = 3 - (\ln 2)n$, treating n as a continuous variable. From the plot, what sequence β_n would you guess is appropriate here: $a_n - a = O(\beta_n)$?
Note: The function for the natural logarithm in matlab is `log`.